

Question

A is a common element. **B** is a disodium salt of a common oxyacid of **A**, containing water of crystallization, and the formula weight of **B** does not exceed 300. When **B** is decomposed by heating, it loses 36.3% of its weight, corresponding to the loss of all water of crystallization. When an appropriate amount of Cl_2 is passed into the $NaOH$ solution of **B**, concentrated and crystallized, a sodium salt crystal **C** is obtained. If excess Cl_2 is passed into the solution and then concentrated, a crystal **D** containing **A** is obtained. **C** loses 14.9% of its weight at $130^\circ C$ and transforms into **E**. The anion of **E** has two lengths of covalent bonds. **E** is further heated to $300^\circ C$ and transforms into **F**, and the anion of **F** has only one length of covalent bond.

The following are several propositions about the above elements and substances:

1. The element **A** exists as a liquid in its elemental form at room temperature and pressure.
2. **B** has strong oxidizing properties.
3. When **B** is converted to **D**, the theoretical molar ratio of **B** to Cl_2 is 1 : 1.
4. The most stable conformation of the anion of **C** does not have a center of symmetry.
5. The process of converting **E** to **F** produces a colorless and odorless gas.
6. The most stable conformation of the anion in **F** contains a center of symmetry.
7. In the anions of **B**, **C**, **D**, **F** respectively, all elements have only one chemical environment.
8. The formula weight of the anion in **B** is greater than the formula weight of the anion in **D**.

Given that the number of true propositions is not a prime number, select the correct option below.

A. All other options are incorrect

B. The number of incorrect descriptions is even.

- C.** The product of the indices of all error descriptions does not exceed 2000
- D.** The sum of the indices of all correctly described items does not exceed 8.
- E.** Among the indices of correct descriptions, there are more odd numbers than even numbers.
- F.** The difference between the largest index of incorrect descriptions and the smallest index of correct descriptions is an even number.

Answer

Correct Answer: **D**

Detailed Explanation

Let the chemical formula of **B** be $Na_2A_xO_y \cdot zH_2O$, then the formula weight $M(\mathbf{B}) = 18.02z/36.3\% = 49.6z$, $M(A_xO_y) = 31.6z - 46$.

CHECKPOINT

1.5 PTS

$$M(A_xO_y) = 31.6z - 46$$

Since $M(\mathbf{B}) < 300$, then $2 \leq z \leq 6$.

CHECKPOINT

1 PTS

$$2 \leq z \leq 6$$

Trying in sequence, we find that when $z = 4$, **B** has a reasonable solution $Na_2SO_3 \cdot 4H_2O$, and when $z = 5$, it has a reasonable solution $Na_2S_2O_3 \cdot 5H_2O$. There are no other reasonable solutions.

CHECKPOINT

0.5 PTS

A is *S*

CHECKPOINT

5 PTS

B may be $Na_2SO_3 \cdot 4H_2O$ or $Na_2S_2O_3 \cdot 5H_2O$

Regardless of which of the above two cases **B** is, it can be known that **D** is $Na_2SO_4 \cdot 10H_2O$. The weight loss of **C** to **E** is also a water loss process. Combined with calculation, **E** is $Na_2S_2O_6$, so **C** is $Na_2S_2O_6 \cdot 2H_2O$. **F** is Na_2SO_4 .

CHECKPOINT

0.5 PTS

D is $Na_2SO_4 \cdot 10H_2O$

CHECKPOINT

1 PTS

C is $Na_2S_2O_6 \cdot 2H_2O$

CHECKPOINT

2 PTS

E is $Na_2S_2O_6$

CHECKPOINT

1 PTS

F is Na_2SO_4

If **B** is $Na_2SO_3 \cdot 4H_2O$, then the correct descriptions are 3 and 7, which contradicts "the number of correct descriptions is not a prime number". If **B** is $Na_2S_2O_3 \cdot 5H_2O$, then only 8 is the correct description. Therefore, **B** is $Na_2S_2O_3 \cdot 5H_2O$.

CHECKPOINT

2 PTS

B is $Na_2S_2O_3 \cdot 5H_2O$

CHECKPOINT

1 PTS

The correct description is 8

Therefore, the correct answer is D.